KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY, KUMASI

INSTITUTE OF DISTANCE LEARNING

DEPARTMENT OF MECHANICAL ENGINEERING

B.Sc. (Eng.)
TAKE HOME MID-SEMESTER EXAMINATION, 2013
Third Year

ME: 366 HEAT TRANSFER

SUBMISSION DATE: 9TH MARCH 2013

INSTRUCTIONS TO CANDIDATES

(a) **Answer all the TWENTY (20) multiple choice questions** IN SECTION A AND TWO (2) QUESTIONS from SECTION B ON THE QUESTION PAPER.

(b) IN SECTION A YOU are required to Select or provide an answer to a question only when you have shown clearly and visibly and *beyond all reasonable doubt* that you have worked towards, and obtained, that very answer. **Do all rough work in PEN on the question paper and not anywhere.**

(c) Provide your Index number and Facilitation centre in the spaces below.

FACILITATION CENTRE: ----------

CANDIDATE'S INDEX NUMBER: ----------

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SECTION A [40 POINTS]

The Stefan-Boltzmann's constant is 5.67×10^{-8} W/ (m^2 K^4) and $g = 9.81$ m/s^2

For the multiple choice questions, circle in pen the best option or supply the correct answer if the answer is not provided. For the other questions, follow specific instructions and answer the questions appropriately. Show working and do all rough work on the question paper.

Q1 – Q5 By considering the steady-state conditions for one-dimensional conduction in a plane as shown in Figure 1, having thermal conductivity $\lambda = 50\,\mathrm{W/m\,K}$ and thickness L = 0.25 m with no internal heat generation, determine the unknown quantities and complete Table 1 below for each case.

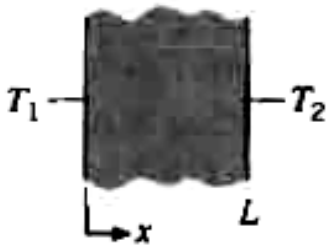


Figure 1

Table 1: Unknown quantities to be determined

Case	T_1/ (^{0}C)	T_2/ (^{0}C)	[dT/dx]/ (K/m)	$\dot{q}$ /(W/m^2)
Q1	50	- 20		
Q2	- 30	- 10		
Q3	70		160	
Q4		40	- 80	
Q5		30	200	

7. The inner and outer surfaces of a 5-m x 6-m brick wall of thickness 30 cm and constant thermal conductivity 1.265 W/ (m ^{0}C) are maintained at temperatures of 35 ^{0}C and 20 ^{0}C, respectively. The rate of heat transfer through the wall, in W, is

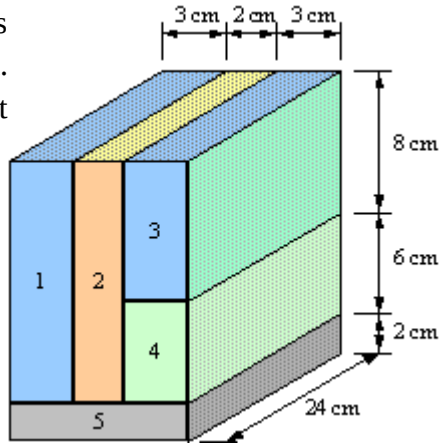
A. 1035
B. 1242
C. 1380
D. 1898
E. 2415

8. A brick wall of thickness 20 cm and thermal conductivity 0.7 W/ (m °C) is maintained at 25 °C at one surface and 4 °C at the other surface. The heat flow rate (**in Watt**) across a 4-m^2 surface area of the wall is
   A. 138
   B. 294
   C. 2.94
   D. 1.38

9. A fibre-glass insulating board with thermal conductivity 0.08 W/ (m °C) is to be used to limit the heat losses to 80 W/m^2 for a temperature difference of 100°C across the board. The thickness (**in mm**) of the insulating board is:
   A. 100
   B. 75
   C. 0.1
   D. 0.075

Use the Information below to answer questions 10-15

For the composite wall shown on the right, thermal conductivities are: $\lambda_1 = \lambda_3 = 80$ W/ (m. K), $\lambda_2 = 120$ W/ (m. K), $\lambda_4 = 100$ W/ (m. K), and $\lambda_5 = 150$ W/ (m. K). Air temperature is 30 ºC on the left side and 80 ºC on the right side. Convection coefficient is $\alpha = 15$ W/ (m^2. K) on both sides.



10. The thermal resistance in (**K/W**) for layer 1 is
    A. 0.01008
    B. 0.01116
    C. 0.02083
    D. 0.02120
    E. 0.1111

11. The thermal resistance in (**K/W**) for layer 4 is
    A. 0.01008
    B. 0.01116
    C. 0.02083
    D. 0.02120
    E. 0.1111

12. The thermal resistance in (**K/W**) for layer 5 is
    A. 0.01008
    B. 0.01116
    C. 0.02083
    D. 0.02120
    E. 0.1111

13. The equivalent thermal resistance in (**K/W**) for layers 3 and 4 in combination is
    A. 0.01008
    B. 0.01116
    C. 0.02083
    D. 0.02120
    E. 0.1111

14. The total thermal resistance for the composite wall in (**K/W**) is
    A. 0.01008
    B. 0.01116
    C. 0.02083
    D. 0.02120
    E. 0.1111

15. The rate of heat transfer in (**W**) through the composite wall is
    A. 14.31
    B. 450.05
    C. 2358.49
    D. 4480.29
    E. 4960.32

16. For heat conduction, the heat transfer rate equation is known as:
    A. Kirchoff's Law
    B. Fourier's Law
    C. Newton's Law
    D. Stefan –Boltzmann's Law
    E. Reciprocity Rule

17. Consider a person standing in a breezy room at a temperature of 20 ^{0}C. The exposed surface area and the average outer surface temperature of this person are 1.6 m^2 and 29 ^{0}C, respectively. The convection heat transfer coefficient and the emissivity of the person are 6 W/m^2 ^{0}C and 0.95, respectively. The total rate of heat transfer from this person, in W, is

A. 81.5 W
B. 81.7 W
C. 86.7 W
D. 163.2 W
E. 168.1 W

***<u>Conduction with internal Heat generation Application of Fourier's law: Use the information below to answer questions 18 - 20</u>*.**

The temperature distribution across a wall 1 m thick at a certain instant of time is given as

$$T(x) = 900 - 300x - 50x^2,$$

where, T is in degree Celsius and x is in metres. A uniform internal heat generation, $\dot{q} = 1000$ W/m^3, is present in the wall of area 10 m^2 having constant properties ρ = 1600 kg/m^3, λ = 40 W/ (m K), and $c_p = 4$ (kJ/kg.K). Assume that there is one-dimensional heat conduction in the positive x-direction.

18. The rate of heat transfer entering the wall (x = 0) is

A. 160 kW
B. 120 kW
C. 100 kW
D. 90 kW
E. 360 kW

19. The rate of heat transfer leaving the wall (x = 1m) is

A. 160 kW
B. 120 kW
C. 100 kW
D. 90 kW
E. 220 kW

20. The rate of energy storage in the wall is

A. – 30 kW
B. – 40 kW
C. – 10 kW
D. – 15 kW
E. – 50 KW

SECTION B (60 POINTS)

ANSWER **ANY TWO QUESTIONS** WITH THE AID OF APPROPRIATE AND READABLE SKETCHES, WHERE APPLICABLE.

QUESTION 1

Water at 65 ^{0}C flows through a 5 cm bore steel pipe 1cm thick. The surroundings are at 20 ^{0}C. The pipe thermal conductivity is 52 W/ (m. K), and the surface heat transfer coefficients inside and outside are respectively 1136 W/ (m^2. K) and 13 W/ (m^2. K). An insulating layer 3 cm thick of thermal conductivity 0.17 W/ (m. K) is added to the outside of the pipe.
Calculate the rate of heat loss per metre length of the pipe and the surface temperatures.

SOLUTION TO QUESTION 1

SOLUTION TO QUESTION 1 CONTINUED

QUESTION 2

The composite wall of an oven, as shown in **FIGURE Q2**, consists of three materials; two of which are of known thermal conductivity k_A= 20 W/ (m. K) and k_C = 50 W/ (m. K) and known thickness L_A = 0.30 m and L_C = 0.15 m. The third material, B, which is sandwiched between materials A and C is of known thickness L_B =0.15 m but of unknown thermal conductivity. Under steady-state operations, measurements revealed an outer surface temperature $T_{s,\,o}$ = 20 ^{0}C and inner surface temperature $T_{s,\,i}$ = 600 ^{0}C and an oven air temperature T_∞ = 800 ^{0}C. The inside convection heat transfer coefficient, h is known to be 25 W/ (m^2. K). What is the value of k_B in W/(m . K)?

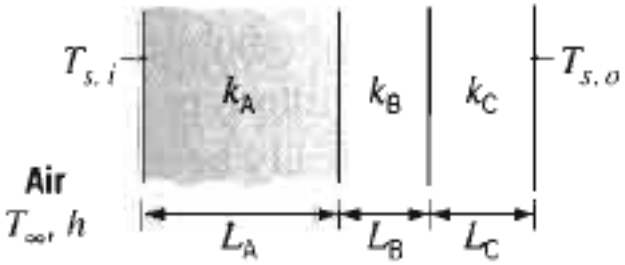


FIGURE Q2.

SOLUTION TO QUESTION 2

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SOLUTION TO QUESTION 2 CONTINUED

QUESTION 3

(a) The inner and outer surfaces of a 0.5-cm-thick 2-m x 2-m window glass in winter are 10 °C and 3 °C, respectively. If the thermal conductivity of the glass is 0.78 W/m·°C, determine the amount of heat loss, in kJ, through the glass over a period of 1 hour.

(b) A 5.0-cm-diameter cylinder is heated to a temperature of 200 °C, and air at 30 °C is forced across it at a velocity of 50 m/s. If the surface emissivity is 0.7, calculate the total heat loss per unit length if the walls of the enclosing room are at 10 °C. Take the convection heat transfer coefficient, α_c as 18 W/ (m^2.°C).

SOLUTION TO QUESTION 3

SOLUTION TO QUESTION 3 CONTINUED